OBSTRUCTION RESIDUES FOR THE $3N-1$ MAP: FULL FACTOR COMPLEXITY, POSITIVE DENSITY, AND A CONSTRUCTIVE LIFT

JAKOB STEMBERGER

Abstract. We study a family $\mathcal{O}_L \subseteq \{1,\dots,2^L\}$ of residue classes modulo 2^L arising naturally in the dynamics of the $3n-1$ map $T_-(n) := (3n-1)/2^{v_2(3n-1)}$. The family is defined by an algebraic criterion on a two-track parallel reduction. We prove three structural results: a constructive lift between modular levels, the existence of a positive asymptotic density c_W with rigorous lower bound $c_W \geq 7556/65536 \approx 0.115$, and full factor complexity $p_W(\ell) = 2^\ell$ of the associated language of binary representations. The central technical tool is a parity lemma in the residue system, which forces the terminal data of the parallel reduction into a shift-s normal form with integer-valued shift index. A finer classification by shift index yields a strict lift balance, an exact series representation of c_W, and reduces the open question of c_W's exact value to a single quantitative bound on a sub-family of atomic obstructions. Results transfer to the classical $3n+1$ map via a residue-level involution $r \mapsto 2^L - r$ that maps the T_- parallel reduction to its T_+ analogue step-by-step.

1. Introduction

1.1. **The Collatz map and its $3n-1$ cousin.** Let $\mathbb{N}_{\mathrm{odd}}$ denote the set of positive odd integers, and let $v_2(m)$ denote the 2-adic valuation of an integer $m \neq 0$. The *Collatz map* T_+ takes an odd integer n to the odd part of $3n+1$:

$$T_+(n) := \frac{3n+1}{2^{v_2(3n+1)}}.$$

The Collatz conjecture, attributed to Lothar Collatz and widely studied since the 1930s, asserts that for every $n \in \mathbb{N}_{\mathrm{odd}}$ the iteration $n, T_+(n), T_+^2(n), \dots$ eventually reaches the fixed point 1. Despite considerable computational evidence and a large body of partial results — early density-side analyses by Terras [Ter76] and Everett [Eve77], refined density estimates by Crandall [Cra78] and Korec [Kor94], and most recently the breakthrough of Tao [Tao22], showing that almost all trajectories attain almost bounded values — the conjecture itself remains open. We refer the reader to the

Date: May 31, 2026.
2020 *Mathematics Subject Classification.* 11B83, 37P05, 37B10, 37B40, 68R15.
Key words and phrases. Collatz conjecture, $3n-1$ map, residue dynamics, factor complexity, subshift, parallel reduction.

annotated bibliographies of Lagarias [Lag10b, Lag10a] for a comprehensive overview of the literature.

The Collatz map has a natural cousin, the $3n-1$ *map* T_- acting on the same domain:

$$T_-(n) := \frac{3n-1}{2^{v_2(3n-1)}}.$$

The two maps are structurally parallel: both perform the operation "multiply by three, add a small constant, strip the factors of two from the result". They differ only in the sign of the additive constant. The trajectories of T_- are, however, *not* believed to converge to a single attractor: three cycles are classically known — the trivial fixed cycle through 1, the $\{5,7\}$-cycle, and the 7-cycle $17 \to 25 \to 37 \to 55 \to 41 \to 61 \to 91 \to 17$; see Lagarias [Lag10a] for historical attribution. The question of whether further cycles exist remains as open as the Collatz conjecture itself. The two maps are connected through a classical identity going back to Lagarias [Lag85]: every T_--trajectory starting from m corresponds, via the substitution $r = 2m+1$, to a single T_+-step followed by a T_+-trajectory. This *Lagarias intertwining* illustrates that any structural feature of T_- has a T_+ counterpart; the explicit transfer we use in Section 7, however, is different — a residue-level involution $r \mapsto 2^L - r$ that conjugates the T_- parallel reduction into the T_+ parallel reduction step-by-step (Lemma 7.2).

The present paper is concerned with a family of residue classes modulo 2^L that arises naturally in the dynamics of T_-. We show that this family — which we shall call the *obstruction residues* for reasons made precise in Section 2 — has three structural properties:

(1) a constructive lift between modular levels L and $L+1$,
(2) positive asymptotic density in $\{1,\dots,2^L\}$ as $L \to \infty$,
(3) maximal combinatorial complexity of its associated language of bit strings.

The third property is in some tension with the first two: a constructive lift and positive density would, in many natural settings, force a certain rigidity of the resulting word language (one might expect a sofic shift or a primitive substitution shift). We prove that no such substitution-type characterization is possible, identifying the obstruction residues as a structurally richer family than the naive analogies would suggest.

1.2. **Parallel reductions and the X-invariant.** The construction is built on a simple observation. For an odd integer r with $v := v_2(r-1) \geq 1$, write $m := (r-1)/2^v$. We run T_- in parallel on r and on m modulo 2^L for a fixed level L, keeping track at each step both of the current values and of their *affine relationship*: a pair $(c_j, d_j) \in \mathbb{Q}^2$ such that $T_-^j(m) \equiv c_j \cdot T_-^j(r) + d_j$ modulo an appropriately reduced power of two. The pair is updated by an explicit rule depending on the 2-adic valuations of the two trajectories. We denote the terminal pair $(c_{\mathrm{end}}, d_{\mathrm{end}})$ and define the scalar invariant

$$X_{\mathrm{end}}(r, L) := 3d_{\mathrm{end}}(r, L) + c_{\mathrm{end}}(r, L).$$

For an odd integer r with $v_2(r-1) \geq 1$ and a level $L > v_2(r-1)$, we call r an *obstruction residue at level* L when $X_{\mathrm{end}}(r, L) = 1$, and write $\mathcal{O}_L \subseteq \{1, \dots, 2^L\}$ for the set of all such residues. Definition 2.2 in Section 2 restates this with the parallel reduction made explicit; the typical case is $v = 1$, i.e. $r \equiv 3 \pmod 4$.

The family $(\mathcal{O}_L)_{L \geq 5}$ is small at first — $\mathcal{O}_5 = \{27\}$, $\mathcal{O}_6 = \{19, 27, 59\}$ — but grows rapidly: direct enumeration gives

$$|\mathcal{O}_5| = 1, \quad |\mathcal{O}_6| = 3, \quad |\mathcal{O}_7| = 8, \quad |\mathcal{O}_8| = 19, \quad \dots, \quad |\mathcal{O}_{16}| = 7556,$$

and the density ratio $|\mathcal{O}_L|/2^L$ rises monotonically. Full counts at all levels $L = 5, \dots, 16$ together with the shift-zero / shift-nonzero and atomic-shift-nonzero decompositions are collected in Table 3 (Appendix A). Three structural questions arise:

(Q1) Does $r_0 \in \mathcal{O}_{L_0}$ guarantee that both r_0 and $r_0 + 2^{L_0}$ lie in $\mathcal{O}_{L_0+1}$?

(Q2) Does $|\mathcal{O}_L|/2^L$ converge to a positive limit, and can the limit be bounded from below?

(Q3) What is the set of binary words appearing as factors of the bit representations of residues in $\bigcup_L \mathcal{O}_L$?

The three questions are not independent: a positive answer to (Q1) gives monotonicity, hence most of (Q2). A constructive form of (Q1), exhibiting arbitrary bit patterns under lift, gives (Q3). All three are answered in the affirmative by the main results; in particular the factor language $\mathcal{F}(\mathcal{O})$ defined below turns out to be all of $\{0,1\}^*$.

1.3. Main results.

Theorem 1.1 (Constructive lift)**.** *Let $r_0 \in \mathcal{O}_{L_0}$. Then both $r_+ := r_0$ and $r_- := r_0 + 2^{L_0}$ are obstruction residues at level $L_0 + 1$.*

The proof, given in Section 3, proceeds by case analysis on a finite invariant of the parallel reduction. (No explicit lower bound on L_0 is needed: for $L_0 \leq 4$ the family $\mathcal{O}_{L_0}$ is empty (Remark 2.3) and the statement is vacuous; the smallest non-vacuous level is $L_0 = 5$ with $\mathcal{O}_5 = \{27\}$. The proof in Section 3 only uses $L_0 > v$, which is automatic from $r_0 \in \mathcal{O}_{L_0}$ by Definition 2.2.) The lift is explicit: for each r_0, the terminal data of the reduction at r_+ or r_- is determined by a closed-form algebraic identity.

Theorem 1.2 (Positive density)**.** *The density ratio $|\mathcal{O}_L|/2^L$ is non-decreasing in L, and the limit*

$$c_W := \lim_{L \to \infty} \frac{|\mathcal{O}_L|}{2^L}$$

exists in $(0, 1]$. For every $L_0 \geq 5$, the rigorous lower bound $c_W \geq |\mathcal{O}_{L_0}|/2^{L_0}$ holds; at $L_0 = 16$ this gives $c_W \geq 7556/65536 \approx 0.115$, sharpening the conservative bound $c_W \geq 3/64 \approx 0.047$ obtained from $L_0 = 6$.

The bound at $L_0 = 16$, using only the algorithmic X_{end}-criterion of Section 2, gives the explicit value $7556/65536 \approx 0.115$ stated above. Raising

L_0 further admits substantial additional improvement — direct enumeration at $L_0 = 32$ already yields $c_W \geq \rho_{32} \approx 0.147$ (Table 1) — but the exact value of c_W remains an open problem. We give in Section 6 a refined decomposition that exhibits c_W as a rigorously convergent series and isolates the remaining open question to a single quantitative bound on a sub-family of obstructions.

For $r \in \mathcal{O}_L$, write $w_r \in \{0,1\}^L$ for the binary representation of r as an L-bit word, least significant bit first. The *factor language* of the obstruction family is

$$\mathcal{F}(\mathcal{O}) := \{\, u \in \{0,1\}^* : u \text{ is a factor of } w_r \text{ for some } r \in \mathcal{O}_L,\ L \geq 5 \,\},$$

and the *factor complexity function* is $p_W(\ell) := |\mathcal{F}(\mathcal{O}) \cap \{0,1\}^\ell|$.

Theorem 1.3 (Full factor complexity). *For every $\ell \geq 0$,*

$$p_W(\ell) = 2^\ell.$$

The proof, in Section 5, is constructive: for each binary word u, we exhibit an explicit residue $r(u)$ in some $\mathcal{O}_L$ having u as a factor. The construction is a direct iteration of the lift theorem.

A consequence of Theorem 1.3 is a sharp structural rigidity (Corollary 5.2): the obstruction family cannot be described as a proper subshift of finite type, a proper sofic shift, or a primitive substitution shift in the sense of Pansiot [Pan84] or Devyatov [Dev15].

1.4. **Strategy.** All three theorems rest on a single algebraic structure: the *shift-s normal form* of the terminal data. We prove in Section 2 (Lemma 2.5) that whenever the parallel reduction terminates with $X_{\text{end}} = 1$, the terminal pair has the form

$$(c_{\text{end}}, d_{\text{end}}) = \left(4^s, \frac{1 - 4^s}{3}\right) \qquad \text{for some } s \in \mathbb{Z}.$$

The integer s, which we call the *shift index*, measures the difference between the 2-adic valuations of the two parallel trajectories. Theorem 1.1 is proved by case analysis on how the shift index evolves under lift; Theorem 1.3 is the iterative consequence; Theorem 1.2 follows by monotonicity from the lift. A refined classification by shift index (Section 6) then re-expresses c_W as a convergent series and isolates the remaining open question.

The shift-s normal form is, to the best of our knowledge, new; in particular, it differs from the Markov-operator setup of Wirsching [Wir98], which works with an invariant probability-density function rather than a terminal algebraic datum. The remaining analysis proceeds by elementary case analysis and combinatorial bookkeeping; the arguments do not depend on specialized Collatz machinery beyond the Lagarias intertwining.

1.5. **Outline.** Section 2 introduces the parallel reduction, defines the X-invariant rigorously, and proves the parity lemma. Section 3 contains the proof of Theorem 1.1, split into the two stop-type cases. Section 4 derives Theorem 1.2 as a direct corollary. Section 5 proves Theorem 1.3 and derives the structural rigidity corollary. Section 6 develops the classification by shift index, proves the strict lift balance and series representation, and isolates the remaining open problem. Section 7 transfers the results to the $3n+1$ side via the residue bijection. Section 8 discusses open questions. Appendix A summarizes computational verifications.

2. The parallel reduction and the X-invariant

2.1. **The basic step.** Recall that for an odd integer n, the map T_- strips the 2-adic part from $3n-1$:

$$T_-(n) = \frac{3n-1}{2^{v_2(3n-1)}}.$$

We need a version of this step that keeps track of the modulus. Fix a level $L \geq 1$, and consider a pair (α, β) with α a positive odd integer and β a power of two, interpreted as a residue α modulo β. A single T_--step on such a pair produces

$$(\alpha', \beta') = \left(\frac{3\alpha - 1}{2^{v_\alpha}}, \frac{3\beta}{2^{v_\alpha}}\right), \qquad v_\alpha := v_2(3\alpha - 1),$$

provided $v_\alpha < v_2(\beta)$. If $v_\alpha \geq v_2(\beta)$, the modulus has been exhausted and the step is undefined; we say the reduction has *terminated*.

Since 3 is odd, $v_2(3\beta) = v_2(\beta)$, so $v_2(\beta') = v_2(\beta) - v_\alpha$: the modulus strictly decreases at each step, and the reduction always terminates after finitely many steps. Iterates of β are in general of the form $3^j \cdot 2^k$ rather than literal powers of two; only the 2-adic valuation $v_2(\beta)$ is used downstream, so we track β as a positive integer of known 2-adic valuation.

2.2. **The two-track setup.** Fix r a positive odd integer with $r \neq 1$, set $v := v_2(r-1)$, and write $m := (r-1)/2^v$; then m is odd. The inequality $v \geq 1$ is automatic for odd $r > 1$ since $r - 1$ is then even. (We use v as a parameter throughout; the special case $v = 1$ — equivalently $r \equiv 3 \pmod 4$ — is the typical reader picture and is presented first wherever a concrete calculation helps.)

Fix a level $L > v$. We run two T_--reductions in parallel:

- the *K-track*, starting from $(a_K^{(0)}, b_K^{(0)}) := (r, 2^L)$,
- the *I-track*, starting from $(a_I^{(0)}, b_I^{(0)}) := (m, 2^{L-v})$.

The initial relation $m = (r-1)/2^v$ writes as $a_I^{(0)} = 2^{-v} a_K^{(0)} - 2^{-v}$. At each step $j \geq 0$, we update both tracks according to the basic step, denoting $v_K^{(j)} := v_2(3a_K^{(j)} - 1)$ and $v_I^{(j)} := v_2(3a_I^{(j)} - 1)$. Cumulative valuations along the two tracks are denoted $V_K^{(j)} := \sum_{i<j} v_K^{(i)}$ and $V_I^{(j)} := \sum_{i<j} v_I^{(i)}$, so that the residual 2-adic valuation of $b_K^{(j)}$ is $L - V_K^{(j)}$ and that of $b_I^{(j)}$ is $L - v - V_I^{(j)}$. The reduction proceeds as long as both tracks can step; it terminates as soon as either condition fails. We write $J = J(r, L)$ for the index of termination.

2.3. **The affine relation and the X-invariant.**

Lemma 2.1. *For each j with $0 \le j \le J$, there exist $c_j, d_j \in \mathbb{Q}$ such that*

$$a_I^{(j)} = c_j\, a_K^{(j)} + d_j \qquad \text{in } \mathbb{Q},$$

with $c_0 = 2^{-v}$, $d_0 = -2^{-v}$, and the update rule

$$c_{j+1} = c_j \cdot 2^{v_K^{(j)} - v_I^{(j)}}, \qquad d_{j+1} = \frac{3d_j + c_j - 1}{2^{v_I^{(j)}}}.$$

Proof. The base case is immediate. Assume the affine relation holds at index j. Writing $A_K^{(j)} := 3a_K^{(j)} - 1$, $A_I^{(j)} := 3a_I^{(j)} - 1$, and $X_j := 3d_j + c_j$,

$$A_I^{(j)} = 3(c_j a_K^{(j)} + d_j) - 1 = c_j(3a_K^{(j)} - 1) + (3d_j + c_j - 1) = c_j A_K^{(j)} + (X_j - 1).$$

Dividing by $2^{v_I^{(j)}}$ and using $A_K^{(j)} = 2^{v_K^{(j)}} a_K^{(j+1)}$:

$$a_I^{(j+1)} = c_j \cdot 2^{v_K^{(j)} - v_I^{(j)}} a_K^{(j+1)} + \frac{X_j - 1}{2^{v_I^{(j)}}},$$

which is the desired affine relation at index $j+1$. $\square$

We define the *X-invariant* of the parallel reduction at (r, L) as

$$X_{\text{end}}(r, L) := X_J = 3d_J + c_J.$$

This is not invariant under the update step; the name reflects its role as the obstruction certificate (Definition 2.2).

2.4. **Definition of obstruction residues.**

Definition 2.2 (Obstruction residue)**.** Let r be a positive odd integer with $r \neq 1$, and write $v := v_2(r-1) \ge 1$. We call r an *obstruction residue at level* $L > v$ if $X_{\text{end}}(r, L) = 1$. We write

$$\mathcal{O}_L := \left\{ r \in \{1, \dots, 2^L\} \;\middle|\; \begin{array}{l} r \text{ odd},\ v_2(r-1) \ge 1, \\ L > v_2(r-1),\ X_{\text{end}}(r, L) = 1 \end{array} \right\}.$$

The definition is algorithmic: given r and L, one runs the parallel reduction, observes the terminal pair (c_J, d_J), and checks $3d_J + c_J = 1$. The computation terminates in at most L steps.

Remark 2.3. A direct enumeration confirms $\mathcal{O}_L = \emptyset$ for $1 \le L \le 4$; the smallest level at which an obstruction residue exists is $L = 5$, with $\mathcal{O}_5 = \{27\}$. At the next level, $\mathcal{O}_6 = \{19, 27, 59\}$, comprising the new atomic residue 19 and the two lifts $\{27, 59\}$ of $27 \in \mathcal{O}_5$.

We call such r *obstruction residues* because $X_{\text{end}}(r, L) = 1$ is exactly the rigidity that forces the two parallel reductions to terminate in lockstep (Corollary 2.6) and collapses their terminal affine relation to the shift-s normal form developed below; a generic residue exhibits no such coupling. The name also has a dynamical reading at the level of T_{-}-trajectories rather than the parallel reduction; this reading motivates the definition but is developed elsewhere, and the present paper uses only the algebraic side established above.

2.5. A worked instance.

Example 2.4 (The reduction at $r = 27$, and a non-obstruction)**.** Take $r = 27$ at level $L = 5$, so $v = v_2(26) = 1$ and $m = 13$. Running the two tracks (with $v_K^{(j)} := v_2(3a_K^{(j)} - 1)$ and $v_I^{(j)} := v_2(3a_I^{(j)} - 1)$, and recording the affine data of Lemma 2.1):

j	$a_K^{(j)}$	$a_I^{(j)}$	$v_K^{(j)}$	$v_I^{(j)}$	c_j	d_j
0	27	13	4	1	$\frac{1}{2}$	$-\frac{1}{2}$
1	5	19	1	3	4	-1

Both tracks stop at $j = 1$: on the K-track $v_K^{(1)} = 1 \geq L - V_K^{(1)} = 5 - 4 = 1$, and on the I-track $v_I^{(1)} = 3 \geq (L - v) - V_I^{(1)} = 4 - 1 = 3$. The terminal pair is $(c_J, d_J) = (4, -1)$, so $X_{\text{end}}(27, 5) = 3 \cdot (-1) + 4 = 1$ and $27 \in \mathcal{O}_5$.

By contrast, $r = 7$ at $L = 5$ (here $v = 1$, $m = 3$) stops at $J = 1$ with terminal pair $(c_J, d_J) = (\frac{1}{4}, -\frac{1}{4})$, giving $X_{\text{end}}(7, 5) = 3\cdot(-\frac{1}{4}) + \frac{1}{4} = -\frac{1}{2} \neq 1$, so $7 \notin \mathcal{O}_5$. The pair pinpoints what the criterion tests: $r = 7$ shares with every obstruction an *even* terminal valuation difference — here $V_K^{(J)} - V_I^{(J)} - v = 2 - 3 - 1 = -2$ — yet fails $X_{\text{end}} = 1$ because its terminal d_J comes out wrong. The parity lemma below shows this even difference is necessary for every obstruction; $r = 7$ shows it is not sufficient.

2.6. The parity lemma.

The central algebraic fact of this paper is the following.

Lemma 2.5 (Parity lemma)**.** *Suppose the parallel reduction at (r, L) terminates with $X_{\text{end}}(r, L) = 1$, and write $v := v_2(r - 1)$. Then*

$$\delta_J := V_K^{(J)} - V_I^{(J)} - v$$

is an even *integer (possibly negative) with $\delta_J \in [-(L - 1), L - 1 - v]$.*

Proof. Iterating the update rule from Lemma 2.1 with $c_0 = 2^{-v}$:

$$c_j = 2^{-v} \cdot \prod_{i<j} 2^{v_K^{(i)} - v_I^{(i)}} = 2^{V_K^{(j)} - V_I^{(j)} - v} = 2^{\delta_j}.$$

From $X_J = 3d_J + c_J = 1$, $d_J = (1 - c_J)/3 = (1 - 2^{\delta_J})/3$. The affine relation at index J becomes

$$a_I^{(J)} = 2^{\delta_J} a_K^{(J)} + \frac{1 - 2^{\delta_J}}{3} \qquad \text{in } \mathbb{Q}.$$

Multiplying by 3:

$$3a_I^{(J)} = 3 \cdot 2^{\delta_J} a_K^{(J)} + 1 - 2^{\delta_J}. \tag{1}$$

We split by sign of δ_J. Write $\delta := \delta_J$ throughout.

Case $\delta \geq 0$. All terms in (1) are integers. Reducing modulo 3: $3a_I^{(J)} \equiv 0$ and $3 \cdot 2^\delta a_K^{(J)} \equiv 0$, so $1 - 2^\delta \equiv 0 \pmod 3$, i.e. $2^\delta \equiv 1 \pmod 3$. Since $2 \equiv -1 \pmod 3$, this forces δ even.

Case $\delta < 0$. Write $\delta = -e$, $e > 0$. Multiplying (1) by 2^e:

$$3 \cdot 2^e\, a_I^{(J)} = 3\, a_K^{(J)} + 2^e - 1.$$

All terms are integers; reducing modulo 3 yields $2^e - 1 \equiv 0 \pmod 3$, hence e even, hence δ even.

In both cases δ is even. For the bound, each firing step satisfies $v_K^{(j)} < L - V_K^{(j)}$ and $v_I^{(j)} < L - v - V_I^{(j)}$ (else the respective stop would have fired at j), so $V_K^{(J)} \leq L - 1$ and $V_I^{(J)} \leq L - 1 - v$; hence $\delta_J = V_K^{(J)} - V_I^{(J)} - v \in [-(L-1), L-1-v]$. $\square$

2.7. **The shift-s normal form.** Writing $\delta_J = 2s$ with $s \in \mathbb{Z}$, the parity lemma yields

$$c_J = 4^s, \qquad d_J = \frac{1 - 4^s}{3}.$$

We call this the *shift-s normal form* and s the *shift index* of r at level L, denoted $s(r, L)$. In particular, $V_K^{(J)} - V_I^{(J)} = 2s + v$. The obstruction family partitions:

$$\mathcal{O}_L = \bigsqcup_{s \in \mathbb{Z}} \mathcal{O}_L^{G_s}, \qquad \mathcal{O}_L^{G_s} := \{\, r \in \mathcal{O}_L : s(r, L) = s \,\}.$$

For $s = 0$ we have $(c_J, d_J) = (1, 0)$, meaning $a_I^{(J)} = a_K^{(J)}$. By the bound $\delta_J \in [-(L-1), L-1-v]$ from Lemma 2.5, $s = \delta_J/2 \in [-(L-1)/2,\, (L-1-v)/2]$; in particular $|s| \leq \lfloor (L-1)/2 \rfloor$ (using that s is an integer).

2.8. **Simultaneous stop and stop types.**

Corollary 2.6 (Simultaneous stop)**.** *If $X_{\mathrm{end}}(r, L) = 1$, then the K-track and I-track satisfy the termination condition $v_\alpha \geq v_2(\beta)$ at the same index J. Moreover, the terminal step valuations satisfy $v_I^{(J)} = v_K^{(J)} + 2s$, where s is the shift index of r at level L from the shift-s normal form above.*

Proof. From the shift-s normal form, $a_I^{(J)} = 4^s a_K^{(J)} + (1 - 4^s)/3$, so $A_I^{(J)} = 4^s A_K^{(J)}$ and hence $v_I^{(J)} = v_K^{(J)} + 2s$. Combining with $v_2(b_I^{(J)}) = (L - v) - V_I^{(J)}$ and $v_2(b_K^{(J)}) = L - V_K^{(J)}$, and using $V_K - V_I = 2s + v$:

$$\begin{aligned} v_I^{(J)} \geq v_2(b_I^{(J)}) &\iff v_K^{(J)} + 2s \geq L - v - V_I^{(J)} \\ &\iff v_K^{(J)} \geq L - V_K^{(J)} \\ &\iff v_K^{(J)} \geq v_2(b_K^{(J)}). \end{aligned}$$

$\square$

Definition 2.7 (Stop type)**.** The *stop type* at (r, L) is *EE* (equality) if $v_K^{(J)} = v_2(b_K^{(J)})$ and $v_I^{(J)} = v_2(b_I^{(J)})$, and *SS* (strict) if $v_K^{(J)} > v_2(b_K^{(J)})$ and

$v_I^{(J)} > v_2(b_I^{(J)})$. By Corollary 2.6 these two cases are exhaustive: a K-track equality forces an I-track equality, and a K-track strict inequality forces an I-track strict inequality.

2.9. **The parameter v and the typical case $v = 1$.** The constructions above are stated for general $v := v_2(r-1) \geq 1$. The typical case is $v = 1$ (equivalently $r \equiv 3 \pmod 4$); for $r \equiv 1 \pmod 4$ the value of v is ≥ 2. In every formula above, setting $v = 1$ recovers the original initial pair $(c_0, d_0) = (\frac{1}{2}, -\frac{1}{2})$ and the parity relation $V_K^{(J)} - V_I^{(J)} \equiv 1 \pmod 2$. The shift-$s$ normal form $(c_J, d_J) = (4^s, (1-4^s)/3)$ is independent of v. The subsequent sections work with the general case throughout; explicit calculations use $v = 1$ wherever this aids the reader.

The $v \geq 2$ residues contribute a small but non-negligible share of $\mathcal{O}_L$: direct enumeration gives, for $L = 16$, $|\mathcal{O}_L| = 7556$ in total, of which 6130 have $v = 1$ and the remaining 1426 (roughly 19%) have $v \geq 2$. The numbers in the appendix table count all v.

3. The lift theorem

We now prove Theorem 1.1: for every $r_0 \in \mathcal{O}_{L_0}$, both lifts $r_+ := r_0$ and $r_- := r_0 + 2^{L_0}$ are obstructions at level $L_0 + 1$. The two halves are stated and proved separately as Theorems 3.1 and 3.2; together they constitute Theorem 1.1. Throughout, $s \in \mathbb{Z}$ is the shift index of r_0 at level L_0, and we use the synchronization $v_I^{(J)} = v_K^{(J)} + 2s$ (Corollary 2.6) freely.

3.1. The r_+ lift.

Theorem 3.1. *Let $r_0 \in \mathcal{O}_{L_0}^{G_s}$. Then $r_+ := r_0 \in \mathcal{O}_{L_0+1}$.*

Proof. Set $v := v_2(r_0 - 1) \geq 1$. The starting pairs at $(r_+, L_0 + 1)$ are $(r_0, 2^{L_0+1})$ and $(m_0, 2^{L_0+1-v})$ where $m_0 = (r_0 - 1)/2^v$. These differ from the corresponding pairs at (r_0, L_0) only in the modulus, which is doubled. Since each basic step depends only on $a_K^{(j)}, a_I^{(j)}$ and not on the modulus (as long as the latter does not constrain the step), the parallel reduction at $(r_+, L_0 + 1)$ produces identical $v_K^{(j)}, v_I^{(j)}, c_j, d_j$ for $j \leq J$ to those at (r_0, L_0).

Stop type EE at L_0. Here $v_K^{(J)} = L_0 - V_K^{(J)}$ exactly. At level $L_0 + 1$, the K-track modulus at index J has 2-adic valuation $L_0 + 1 - V_K^{(J)} = v_K^{(J)} + 1 > v_K^{(J)}$, so the step is permitted. Taking one more step, with $c_J = 4^s$, $d_J = (1-4^s)/3$:

$$c_{J+1} = 4^s \cdot 2^{v_K^{(J)} - v_I^{(J)}} = 4^s \cdot 2^{-2s} = 1,$$

$$d_{J+1} = \frac{3(1-4^s)/3 + 4^s - 1}{2^{v_I^{(J)}}} = 0.$$

After this step, $V_K^{(J+1)} = L_0$, so the K-track modulus has valuation 1, and the next step would require $v_K^{(J+1)} < 1$. Since $a_K^{(J+1)}$ is odd, $3a_K^{(J+1)} - 1$ is even, hence $v_K^{(J+1)} \geq 1$ — so the next step is impossible. Termination occurs at index $J+1$ with $(c_{J+1}, d_{J+1}) = (1,0)$, hence $X_{\text{end}} = 1$ and $r_+ \in \mathcal{O}^{G_0}_{L_0+1}$.

Stop type SS at L_0. Here $v_K^{(J)} > L_0 - V_K^{(J)}$, i.e. $v_K^{(J)} \geq L_0+1-V_K^{(J)}$ (integer), which is exactly the termination condition at level L_0+1. Termination at (r_+, L_0+1) occurs at index J with terminal data unchanged, so $r_+ \in \mathcal{O}^{G_s}_{L_0+1}$. ☐

3.2. **The r_- lift.**

Theorem 3.2. *Let $r_0 \in \mathcal{O}^{G_s}_{L_0}$. Then $r_- := r_0 + 2^{L_0} \in \mathcal{O}_{L_0+1}$.*

Proof. Write $a^{(j)}_{K,+}, a^{(j)}_{I,+}$ for the values at (r_+, L_0+1) (from Theorem 3.1) and $a^{(j)}_{K,-}, a^{(j)}_{I,-}$ for those at (r_-, L_0+1). The discrepancies

$$\Delta_K^{(j)} := a^{(j)}_{K,-} - a^{(j)}_{K,+}, \qquad \Delta_I^{(j)} := a^{(j)}_{I,-} - a^{(j)}_{I,+}$$

satisfy $\Delta_K^{(0)} = 2^{L_0}$, $\Delta_I^{(0)} = 2^{L_0-v}$ (the I-track shift $\Delta_I^{(0)} = (r_- - 1)/2^v - m_0 = 2^{L_0}/2^v$ uses $v_2(r_- - 1) = \min(v_2(r_0-1), L_0) = v$, valid for $L_0 > v$). We claim that as long as $V_K^{(j)} < L_0$ and $V_I^{(j)} < L_0 - v$, the two reductions take identical local steps,

$$v_2\Big(3a^{(j)}_{K,-} - 1\Big) = v_2\Big(3a^{(j)}_{K,+} - 1\Big) = v_K^{(j)} \tag{2}$$

(and analogously for the I-track), and the discrepancies have the closed form

$$\Delta_K^{(j)} = 2^{L_0 - V_K^{(j)}} \cdot 3^j, \qquad \Delta_I^{(j)} = 2^{L_0 - v - V_I^{(j)}} \cdot 3^j. \tag{3}$$

We prove (2) and (3) jointly by induction on j in the range $0 \leq j < J$, where J is the termination index of the reduction at (r_0, L_0). The base case $j=0$ is immediate. For the inductive step, assume (3) at step j and $j < J$. Since the K-track at (r_0, L_0) has not yet terminated, $v_K^{(j)} < L_0 - V_K^{(j)} = v_2(\Delta_K^{(j)})$. The decomposition $3a^{(j)}_{K,-} - 1 = (3a^{(j)}_{K,+} - 1) + 3\Delta_K^{(j)}$ then has its two summands at strictly different valuations $v_K^{(j)}$ and $L_0 - V_K^{(j)}$, so by the ultrametric identity $v_2(x+y) = \min(v_2x, v_2y)$ when $v_2x \neq v_2y$, the sum has valuation exactly $v_K^{(j)}$. This proves (2) at step j. The basic step then yields $\Delta_K^{(j+1)} = (3\Delta_K^{(j)})/2^{v_K^{(j)}} = 2^{L_0 - V_K^{(j+1)}} \cdot 3^{j+1}$; the same calculation on the I-track completes (3) at step $j+1$.

Stop type SS at L_0. At index J of (r_+, L_0+1), $v_K^{(J)} > L_0 - V_K^{(J)}$, hence $v_2(\Delta_K^{(J)}) = L_0 - V_K^{(J)} < v_K^{(J)}$. The relation $3a^{(J)}_{K,-} - 1 = (3a^{(J)}_{K,+} - 1) + 3\Delta_K^{(J)}$ now has the two summands at distinct valuations $v_K^{(J)}$ and $L_0 - V_K^{(J)}$, so by the ultrametric inequality the sum has valuation $v_K^{(J),-} := L_0 - V_K^{(J)} < L_0+1-V_K^{(J)}$. The K-step at (r_-, L_0+1) at index J is therefore permitted. The same calculation on the I-track yields $v_I^{(J),-} := L_0 - v - V_I^{(J)}$, also permitted, and one more step follows. The required strict inequality $v_2(\Delta_I^{(J)}) = L_0 - v -$

$V_I^{(J)} < v_I^{(J)}$ is the I-track version of SS: by the synchronisation $v_I^{(J)} = v_K^{(J)} + 2s$ of Corollary 2.6 together with $V_K^{(J)} - V_I^{(J)} = 2s + v$, the K-track strict inequality $v_K^{(J)} > L_0 - V_K^{(J)}$ is equivalent to $v_I^{(J)} > L_0 - v - V_I^{(J)}$. Using $V_K - V_I = 2s + v$, the new exponent in the update rule is $v_K^{(J),-} - v_I^{(J),-} = -2s$, so

$$c_{J+1} = c_J \cdot 2^{-2s} = 4^s \cdot 4^{-s} = 1, \qquad d_{J+1} = \frac{3 \cdot (1-4^s)/3 + 4^s - 1}{2^{v_I^{(J),-}}} = 0.$$

After this step $V_K^{(J+1)} = L_0$ and $V_I^{(J+1)} = L_0 - v$, so both moduli have valuation 1 and the next step on either track is impossible for odd a_K, a_I. Termination occurs at index $J+1$ with $(c_{J+1}, d_{J+1}) = (1, 0)$, so $r_- \in \mathcal{O}_{L_0+1}^{G_0}$.

Stop type EE at L_0. Here $v_K^{(J)} = L_0 - V_K^{(J)} = v_2(\Delta_K^{(J)})$, so the two summands of $3a_{K,-}^{(J)} - 1 = (3a_{K,+}^{(J)} - 1) + 3\Delta_K^{(J)}$ have *equal* valuation $v_K^{(J)}$. The ultrametric inequality then gives only $v_2(A_{K,-}^{(J)}) \geq v_K^{(J)} + 1$; that is, $v_2(A_{K,-}^{(J)}) \geq L_0 + 1 - V_K^{(J)}$, which is the K-track stop condition at level L_0+1. The same argument on the I-track gives $v_2(A_{I,-}^{(J)}) \geq L_0 + 1 - v - V_I^{(J)}$, the I-track stop condition at level $L_0 + 1$. Termination therefore occurs at index J with terminal data unchanged, so $r_- \in \mathcal{O}_{L_0+1}^{G_s}$. □

3.3. **The lift symmetry.**

Proposition 3.3 (Lift symmetry)**.** *Let $r_0 \in \mathcal{O}_{L_0}^{G_s}$. The shift indices of the two level-(L_0+1) lifts $r_+ := r_0$ and $r_- := r_0 + 2^{L_0}$ depend only on the stop type of r_0 at L_0 (Definition 2.7) as follows (both lifts at level L_0+1):*

Stop type of r_0 at L_0	*shift index of r_+*	*shift index of r_-*
EE	0	s
SS	s	0

Proof. The four entries are read off directly from the case-analysis proofs of Theorems 3.1 and 3.2: each case computes the terminal data of the lift and identifies the resulting shift index. □

So lifting an obstruction with shift index $s \neq 0$ always produces one shift-s obstruction and one shift-zero obstruction at the next level; lifting a shift-zero obstruction produces two shift-zero obstructions (Figure 1). This is the symmetry that drives the recurrence developed in Section 6.

4. Positive density

Proof of Theorem 1.2. By Theorem 1.1, every $r_0 \in \mathcal{O}_{L_0}$ contributes two distinct elements r_0 and $r_0 + 2^{L_0}$ to $\mathcal{O}_{L_0+1}$. Hence $|\mathcal{O}_{L_0+1}| \geq 2|\mathcal{O}_{L_0}|$, equivalently $|\mathcal{O}_{L_0+1}|/2^{L_0+1} \geq |\mathcal{O}_{L_0}|/2^{L_0}$: the density ratio is non-decreasing. Since $\mathcal{O}_L \subseteq \{1, \ldots, 2^L\}$ the ratio is bounded above by 1. By monotone

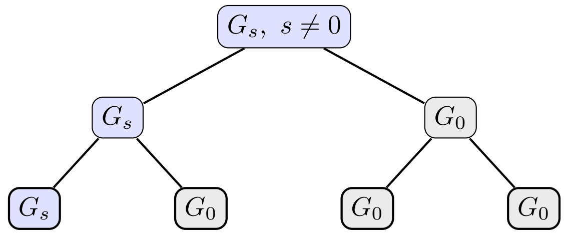

FIGURE 1. Two generations of lifts (Proposition 3.3). A shift-nonzero obstruction (blue) splits into one shift-nonzero and one shift-zero child; a shift-zero obstruction (grey) splits into two shift-zero children. Shift-zero is therefore *absorbing* under lifting: it never produces a shift-nonzero residue. The single persisting shift-nonzero line, shedding one shift-zero child per level, is the combinatorial engine behind the strict lift balance of Corollary 6.4.

convergence, $c_W := \lim_{L\to\infty} |\mathcal{O}_L|/2^L$ exists in $[\rho_{L_0}, 1]$ for every $L_0 \geq 5$. Substituting the direct enumeration $|\mathcal{O}_{16}| = 7556$ (Appendix A, Table 3) yields $c_W \geq 7556/65536 \approx 0.115$; the conservative bound $c_W \geq 3/64$ is recovered at $L_0 = 6$ from $\mathcal{O}_6 = \{19, 27, 59\}$ (Section 2). $\square$

The conservative bound $\rho_6 = 3/64$ uses only the three explicit obstructions $\mathcal{O}_6 = \{19, 27, 59\}$. By raising L_0 and computing $|\mathcal{O}_{L_0}|$ directly, tighter rigorous bounds for c_W follow (Table 1):

L_0	$\lvert\mathcal{O}_{L_0}\rvert$	ρ_{L_0}	rigorous lower bound for c_W
6	3	≈ 0.047	$3/64$
10	91	≈ 0.089	$91/1024$
14	1776	≈ 0.108	$1776/16384$
16	7556	≈ 0.115	$7556/65536$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
32	631538769	≈ 0.147	$631538769/2^{32}$

TABLE 1. Rigorous lower bounds for c_W obtained from finite-level enumeration of $\mathcal{O}_{L_0}$. The ratio $\rho_{L_0} := |\mathcal{O}_{L_0}|/2^{L_0}$ is non-decreasing in L_0 (Theorem 1.2).

These are not asymptotic estimates: they are finite-level computations producing rigorous bounds at the cost of $O(L_0 \cdot 2^{L_0})$ operations to enumerate obstructions modulo 2^{L_0}. The exact value of c_W is not pinned down by this approach; we return to it in Section 6.

5. Full factor complexity

5.1. **Setup.** To each $r \in \mathcal{O}_L$ we associate its binary representation as an L-bit word, least significant bit first:

$$w_r \in \{0,1\}^L, \qquad w_r[k] := \lfloor r/2^k \rfloor \bmod 2, \quad k = 0, \dots, L-1.$$

In particular $w_r[0] = 1$ (since r is odd) and $w_r[L-1]$ is the leading bit of r in this L-bit representation. The *language of obstruction words* at level L is $\mathcal{F}_L := \{w_r : r \in \mathcal{O}_L\}$, and the *factor language* of the obstruction family is

$$\mathcal{F}(\mathcal{O}) := \{\, u \in \{0,1\}^* : u \text{ is a factor of } w_r \text{ for some } r \in \mathcal{O}_L,\ L \geq 5\,\}.$$

The *factor complexity function* is $p_W(\ell) := |\mathcal{F}(\mathcal{O}) \cap \{0,1\}^\ell|$. Trivially $p_W(\ell) \leq 2^\ell$.

5.2. **The main theorem.**

Theorem (restating Theorem 1.3)**.** *For every $\ell \geq 0$, $p_W(\ell) = 2^\ell$.*

Proof. The case $\ell = 0$ is the empty word, which is trivially a factor of every w_r, so $p_W(0) = 1 = 2^0$. Fix $\ell \geq 1$ and $u = (u_0, u_1, \dots, u_{\ell-1}) \in \{0,1\}^\ell$. Choose any anchor $r_0 \in \mathcal{O}_6 = \{19, 27, 59\}$ (verified by direct enumeration; see Appendix A). In fact each of the three anchors individually realises every pattern, as direct enumeration via the lift-symmetry table confirms; for the proof one anchor suffices. Define inductively

$$r_k := r_{k-1} + u_{k-1} \cdot 2^{5+k}, \qquad k = 1, 2, \dots, \ell.$$

We claim $r_k \in \mathcal{O}_{6+k}$ for each k. The base $k = 0$ is by choice of r_0. For the inductive step from $k-1$ to k: if $u_{k-1} = 0$ then $r_k = r_{k-1}$, so r_k is the r_+ lift and lies in $\mathcal{O}_{6+k}$ by Theorem 3.1; if $u_{k-1} = 1$ then $r_k = r_{k-1} + 2^{5+k}$ is the r_- lift and lies in $\mathcal{O}_{6+k}$ by Theorem 3.2.

By construction, $w_{r_\ell}[5+k] = u_{k-1}$ for $k = 1, \dots, \ell$, so u appears as a factor of w_{r_ℓ} at positions $6, \dots, 5+\ell$. Thus $u \in \mathcal{F}(\mathcal{O})$ for every $u \in \{0,1\}^\ell$, hence $p_W(\ell) \geq 2^\ell$. Combined with the trivial upper bound, $p_W(\ell) = 2^\ell$. ☐

5.3. **Entropy and substitution-shift rigidity.** The *topological entropy* of a language $\mathcal{L} \subseteq \{0,1\}^*$ is $h(\mathcal{L}) := \limsup_{\ell\to\infty} (1/\ell) \log p_{\mathcal{L}}(\ell)$; the factor-complexity function $p_{\mathcal{L}}$ itself goes back to the foundational symbolic-dynamics papers of Morse–Hedlund [MH38, MH40]. See Lind–Marcus [LM95] for general background on entropy of subshifts and subshifts of finite type, Fogg [Fog02] or Allouche–Shallit [AS03] for factor complexity of substitution and automatic sequences, Cassaigne [Cas97] for the general theory of subword complexity in terms of special factors, and Berthé–Rigo [BR10] and Lothaire [Lot02] for modern collected references on combinatorics on words, automata and number theory.

Corollary 5.1. *The topological entropy of $\mathcal{F}(\mathcal{O})$ is* $\log 2$*.*

This is the maximal value, matching that of the full Bernoulli shift $\{0,1\}^{\mathbb{Z}}$.

Corollary 5.2. *The obstruction family $(\mathcal{O}_L)_{L\geq 5}$, viewed as a graded family of finite languages over $\{0,1\}$, is*

(i) not the language of a proper subshift of finite type, that is, one defined by a non-empty list of forbidden factors;
(ii) not the language of a proper sofic shift in the classical sense (image of a proper SFT under a one-block factor map);
(iii) not the factor language of a primitive substitution shift in the sense of Pansiot [Pan84] *or Devyatov* [Dev15]*.*

Proof. A non-trivial SFT defined by a non-empty finite list of forbidden subwords has entropy strictly less than $\log 2$ (Lind–Marcus [LM95]), contradicting Corollary 5.1. A non-trivial sofic shift is the image of an SFT under a one-block factor map and inherits the entropy constraint. Primitive substitution shifts have factor complexity in one of the polynomial classes $\Theta(1)$, $\Theta(n)$, $\Theta(n \log\log n)$, $\Theta(n\log n)$, $\Theta(n^2)$ (Pansiot [Pan84]; refined by Devyatov [Dev15]), and therefore have topological entropy zero. The obstruction family has $p_W(\ell) = 2^\ell$ and hence entropy $\log 2 > 0$, so it cannot be the factor language of a primitive substitution shift. □

Remark 5.3. The restriction to *primitive* substitution shifts in Corollary 5.2(iii) is the scope of the Pansiot–Devyatov classification we invoke. The entropy obstruction of Corollary 5.1 in fact excludes the entire zero-entropy regime, including all linearly recurrent subshifts, which contain the primitive substitution shifts [DHS99]; the substitutive candidates it does *not* reach are therefore the non-primitive shifts of positive entropy. Whether $\mathcal{F}(\mathcal{O})$ can be realised as the factor language of such a shift—or, more generally, within the S-adic framework [BD14], which builds sequences by composing morphisms from a possibly infinite set and so goes beyond single primitive substitutions—remains open (see Section 8).

6. The G_s classification

6.1. **Atomic obstructions.** An obstruction $r \in \mathcal{O}_L$ is *atomic at level L* if $r \bmod 2^{L-1} \notin \mathcal{O}_{L-1}$. We write $\mathcal{A}_L \subseteq \mathcal{O}_L$ and $\mathcal{A}_L^{G_s} := \mathcal{A}_L \cap \mathcal{O}_L^{G_s}$.

Lemma 6.1 (Atom decomposition)**.** *For every $L \geq 5$,*

$$|\mathcal{O}_L| = \sum_{L_0=5}^{L} |\mathcal{A}_{L_0}| \cdot 2^{L-L_0}.$$

Proof. We show that every $r \in \mathcal{O}_L$ has a unique smallest $L_0 \geq 5$ with $r \bmod 2^{L_0} \in \mathcal{A}_{L_0}$. *Existence:* starting from r at level L, consider the descending sequence $r_k := r \bmod 2^{L-k}$ for $k = 0, 1, \ldots, L-5$. By the lift theorem (Theorem 1.1), whenever $r_{k+1} \in \mathcal{O}_{L-k-1}$, the level-$(L-k)$ residue $r_k = r \bmod 2^{L-k}$ — being one of the two lifts of r_{k+1} at level $L-k$ — lies in $\mathcal{O}_{L-k}$; the converse need not hold. Let L_0 be the smallest level ≥ 5 for which $r \bmod 2^{L_0} \in \mathcal{O}_{L_0}$. Then $r \bmod 2^{L_0-1} \notin \mathcal{O}_{L_0-1}$ (either because $L_0 = 5$ and

$\mathcal{O}_4 = \emptyset$ by Remark 2.3, or by minimality), so $r \bmod 2^{L_0} \in \mathcal{A}_{L_0}$. *Uniqueness:* the level L_0 is the smallest such by definition, hence unique. Given the atomic anchor $r_0 := r \bmod 2^{L_0}$, the residue r is determined by r_0 together with the bits $\{u_{L_0}, \ldots, u_{L-1}\}$ encoding the lift choices (Theorems 3.1 and 3.2). Each atomic anchor at level L_0 thus generates 2^{L-L_0} obstructions at level L. $\square$

6.2. **No shift-zero atoms.**

Theorem 6.2. *For every $L \geq 7$, $|\mathcal{A}_L^{G_0}| = 0$.*

The conclusion also holds at the base levels $L = 5, 6$ (see Corollary 6.3); the bound $L \geq 7$ in the statement is conservative. At $L = 5$ it is vacuous, since $|\mathcal{O}_5^{G_0}| = 0$; at $L = 6$ the unique shift-zero obstruction is $r = 27$, whose reduction terminates at $J = 2$ with $V_K^{(J)} = 5 = L - 1$, so it falls in Case B below and the case analysis applies. We discharge both base cases by direct enumeration in Corollary 6.3.

Proof. Fix $L \geq 7$ and suppose $r \in \mathcal{O}_L^{G_0}$, so $X_{\mathrm{end}}(r, L) = 1$ with shift index $s = 0$, i.e. $(c_J, d_J) = (1, 0)$. Write $r' := r \bmod 2^{L-1}$. We first note $J \geq 2$: at $J = 0$ the terminal pair is $(2^{-v}, -2^{-v})$, whose second coordinate -2^{-v} is nonzero for every v, contradicting $d_J = 0$; at $J = 1$ the update rule gives $d_1 = (3 \cdot (-2^{-v}) + 2^{-v} - 1)/2^{v_I^{(0)}} = -(1 + 2^{1-v})/2^{v_I^{(0)}}$, whose numerator is strictly negative for $v \geq 1$ (since $2^{1-v} + 1 > 0$), so $d_1 \neq 0$ and again contradicting $d_J = 0$. Hence $J \geq 2$, so the case distinction below over the indices $J-1$ and J is well-defined. The parallel reduction at $(r', L-1)$ starts from the same residue class $a_K \bmod 2^{L-1}, a_I \bmod 2^{L-1-v}$ as at (r, L), with the modulus reduced by one power of two; the two K-tracks (resp. I-tracks) differ as integers by $\Delta_K^{(j)} = 2^{L-1-V_K^{(j)}} \cdot 3^j$ (resp. $\Delta_I^{(j)} = 2^{L-1-v-V_I^{(j)}} \cdot 3^j$), in direct analogy with the lift-minus calculation of Theorem 3.2. Whenever $v_K^{(j)} < L - 1 - V_K^{(j)}$, the ultrametric identity forces $v_2(3a_K^{(j)} - 1)$ to agree at (r, L) and $(r', L-1)$, so the per-step valuations $v_K^{(j)}, v_I^{(j)}$ and hence the affine data (c_j, d_j) coincide between the two reductions until the level-$(L-1)$ stop fires.

Termination of the reduction at (r, L) at index J means at least one track satisfies its stop condition there; by Corollary 2.6, both do. Since $s = 0$, at the terminal index J the synchronisation $v_I^{(J)} = v_K^{(J)} + 2s = v_K^{(J)}$ and $V_K^{(J)} - V_I^{(J)} = v$ holds (Lemma 2.5), so the K-track stop condition $v_K^{(J)} \geq L - V_K^{(J)}$ and the I-track stop condition $v_I^{(J)} \geq (L - v) - V_I^{(J)}$ at $j = J$ are equivalent: substituting $V_I^{(J)} = V_K^{(J)} - v$ and $v_I^{(J)} = v_K^{(J)}$ into the I-track condition reduces it to $v_K^{(J)} \geq L - V_K^{(J)}$, so both stop conditions at $j = J$ read $V_K^{(J)} + v_K^{(J)} \geq L$. (The K- and I-track stop conditions are not in general equivalent at $j < J$, so we rule out child stops on each track separately below; the equivalence is used only at the terminal index J, where it follows from $s = 0$.) We split by the K-track value $V_K^{(J)}$ (equivalently $V_K^{(J-1)} + v_K^{(J-1)}$):

either $V_K^{(J)} < L-1$ (Case A) or $V_K^{(J)} \geq L-1$ (Case B); these exhaust the integer possibilities. We first rule out the K-track stop at level $L-1$ firing at any earlier index $J' \leq J-2$: in that case $V_K^{(J')} + v_K^{(J')} \geq L-1$, the (r, L) reduction takes its J'-th step (the level-L stop has not yet fired since $J' < J$), giving $V_K^{(J'+1)} \geq L-1$, so the K-track modulus at level L at index $J'+1$ has valuation $L - V_K^{(J'+1)} \leq 1$. Since $a_K^{(J'+1)}$ is odd, $v_K^{(J'+1)} \geq 1$, so the next step is impossible: the (r, L) reduction would stop at $J'+1 \leq J-1$, contradicting the definition of J. The same argument applied to the I-track rules out the child I-track stop firing at any index $J' \leq J-2$: an I-stop at J' would force $V_I^{(J'+1)} \geq L-1-v$, so the I-track modulus at level L at index $J'+1$ would have valuation $L - v - V_I^{(J'+1)} \leq 1$ and $v_I^{(J'+1)} \geq 1$ (since $a_I^{(J'+1)}$ is odd) would terminate the parent at $J'+1 \leq J-1$, again contradicting the definition of J. At index $J-1$, a child I-stop requires $V_I^{(J)} \geq L-1-v$; by the terminal synchronisation $V_I^{(J)} = V_K^{(J)} - v$ (from $s = 0$, equivalently $\delta_J = 0$), this is exactly $V_K^{(J)} \geq L-1$ — the Case B branch below — so in Case A neither child stop fires before index J.

Case A: $V_K^{(J)} < L-1$. The K-track stop at level $L-1$ does not fire at any index $\leq J-1$ (the indices $J' \leq J-2$ are ruled out by the preceding paragraph; the index $J' = J-1$ would force $V_K^{(J)} = V_K^{(J-1)} + v_K^{(J-1)} \geq L-1$, contradicting the Case-A hypothesis $V_K^{(J)} < L-1$), so the $(r', L-1)$ reduction runs at least as far as index J; at $j = J$ we have $V_K^{(J)} + v_K^{(J)} \geq L > L-1$, hence the K-track stop at level $L-1$ fires at J and the reduction terminates there. (The I-track stop at level $L-1$ also fires at J by the same chain — at $j = J$ the synchronisation $v_I^{(J)} = v_K^{(J)}$, $V_K^{(J)} - V_I^{(J)} = v$ holds — but only the first track to stop matters for termination.) The terminal data of the two reductions coincide: $(c_J, d_J) = (1, 0)$, so $r' \in \mathcal{O}_{L-1}^{G_0}$ and r is not atomic.

Case B: $V_K^{(J)} \geq L-1$, *equivalently* $V_K^{(J-1)} + v_K^{(J-1)} \geq L-1$. Combined with the standing $V_K^{(J-1)} + v_K^{(J-1)} < L$ (since the level-L stop does not fire at $J-1$), the only integer value is $V_K^{(J)} = L-1$, so the K-track stop at level $L-1$ fires at $J-1$ and the $(r', L-1)$ reduction terminates one step earlier, at index $J-1$, with terminal data (c_{J-1}, d_{J-1}) (whether the I-track stop at level $L-1$ also fires at $J-1$ is immaterial for termination). From the update rule (Lemma 2.1) at index $J-1$,

$$c_J = c_{J-1} \cdot 2^{v_K^{(J-1)} - v_I^{(J-1)}}, \qquad d_J = \frac{3d_{J-1} + c_{J-1} - 1}{2^{v_I^{(J-1)}}}.$$

Setting $c_J = 1$, $d_J = 0$ and reading off the numerator of d_J: $3d_{J-1} + c_{J-1} - 1 = 0$, i.e. $X_{J-1} = 1$. By the parity lemma applied at the terminal index $J-1$ of the reduction at $(r', L-1)$, the terminal pair is in shift-s' normal form for some $s' \in \mathbb{Z}$, so $r' \in \mathcal{O}_{L-1}^{G_{s'}}$.

In either case, $r' \in \mathcal{O}_{L-1}$, so r is not atomic. $\square$

Corollary 6.3. *For every $L \geq 5$, $|\mathcal{A}_L^{G_0}| = 0$.*

Proof. For $L \geq 7$, this is Theorem 6.2. At the two base levels direct enumeration (verified by `count_obstructions.py`) gives $\mathcal{A}_5 = \{27\}$ and $\mathcal{A}_6 = \{19\}$, each with shift index $s = 1 \neq 0$, so $|\mathcal{A}_5^{G_0}| = |\mathcal{A}_6^{G_0}| = 0$ as well. □

6.3. **Strict lift balance and series representation.** Write $g_L := |\mathcal{O}_L^{G_0}|$ and $h_L := |\mathcal{O}_L^{G\neq 0}|$. The lift symmetry of Proposition 3.3 gives, per parent at $L-1$ (Table 2):

Parent at $L-1$	Contribution to g_L	Contribution to h_L
G_0	2	0
G_s, $s \neq 0$	1	1

TABLE 2. Per-parent contribution of a level-$(L-1)$ obstruction to the level-L shift-zero and shift-nonzero populations under the two lifts $r_\pm$.

Summing and adding the atomic contributions, which by Corollary 6.3 occur only in h_L:

Corollary 6.4 (Strict lift balance)**.** *For every $L \geq 6$,*

$$g_L = 2g_{L-1} + h_{L-1}, \qquad h_L = h_{L-1} + |\mathcal{A}_L^{G\neq 0}|.$$

Adding, $|\mathcal{O}_L| = 2|\mathcal{O}_{L-1}| + |\mathcal{A}_L^{G\neq 0}|$.

Theorem 6.5. *The density constant c_W admits the convergent series representation*

$$c_W = \frac{3}{64} + \sum_{L\geq 7} \frac{|\mathcal{A}_L^{G\neq 0}|}{2^L},$$

with non-negative terms. In particular, $|\mathcal{A}_L^{G\neq 0}|/2^L \to 0$ as $L \to \infty$.

Proof. By Lemma 6.1 every $r \in \mathcal{O}_L$ has a unique atomic anchor at some level $L_0 \in \{5, 6, \ldots, L\}$. By Corollary 6.3, $|\mathcal{A}_{L_0}^{G_0}| = 0$ for every $L_0 \geq 5$, so each atom contributes to $\mathcal{A}_{L_0}^{G\neq 0}$. Direct enumeration at the two base levels (verified by `count_obstructions.py`) gives $\mathcal{A}_5 = \{27\}$ and $\mathcal{A}_6 = \{19\}$, both with shift index $s = 1$, so $|\mathcal{A}_5^{G\neq 0}| = |\mathcal{A}_6^{G\neq 0}| = 1$. Hence for every $L \geq 6$,

$$|\mathcal{O}_L| = 2^{L-5} + 2^{L-6} + \sum_{L_0=7}^{L} |\mathcal{A}_{L_0}^{G\neq 0}| \cdot 2^{L-L_0} = 3 \cdot 2^{L-6} + \sum_{L_0=7}^{L} |\mathcal{A}_{L_0}^{G\neq 0}| \cdot 2^{L-L_0}.$$

Dividing both sides by 2^L,

$$\rho_L = \frac{|\mathcal{O}_L|}{2^L} = \frac{3}{64} + \sum_{L_0=7}^{L} \frac{|\mathcal{A}_{L_0}^{G\neq 0}|}{2^{L_0}}.$$

By Theorem 1.2, ρ_L is non-decreasing in L and bounded above by 1, so its limit c_W exists in $(0,1]$. The right-hand side is the L-th partial sum of a non-negative series with fixed first term $3/64$; since the partial sums converge to c_W, the series converges, with sum c_W and tail $|\mathcal{A}_L^{G\neq 0}|/2^L \to 0$ as $L \to \infty$. □

6.4. **The remaining open question.** Theorem 6.5 reduces the question of c_W's exact value to controlling the growth of $|\mathcal{A}_L^{G\neq 0}|$. A bound of the form $|\mathcal{A}_L^{G\neq 0}| \leq C \cdot \alpha^L$ with $\alpha < 2$ would give a geometrically convergent series with explicit error bounds. We do not establish such a bound here; we return to the question in Section 8.

7. The T_+ side

The classical Collatz map T_+ admits the same obstruction theory. We transfer the framework of Section 2 to T_+ and show that the residue involution $r \mapsto \overline{r} := 2^L - r$ realises an explicit bijection between the two obstruction families, step-by-step at the level of the parallel reduction.

7.1. **The T_+ parallel reduction.** Fix a positive odd integer r with $v_2(r+1) \geq 1$, and write $v := v_2(r+1)$ and $m := (r+1)/2^v$; then m is odd. For a level $L > v$, run the parallel reduction with T_+:

- the *K-track*, starting from $(a_K^{(0)}, b_K^{(0)}) := (r, 2^L)$ with basic step $a'_K = (3a_K+1)/2^{v_K}$, $v_K := v_2(3a_K+1)$;
- the *I-track*, starting from $(a_I^{(0)}, b_I^{(0)}) := (m, 2^{L-v})$ with the analogous basic step.

The initial relation $m = (r+1)/2^v$ writes as $a_I^{(0)} = 2^{-v}a_K^{(0)} + 2^{-v}$, so the affine relation $a_I^{(j)} = c_j a_K^{(j)} + d_j$ holds at index 0 with $c_0 = 2^{-v}$, $d_0 = +2^{-v}$. The derivation of Lemma 2.1, applied with T_+ in place of T_- and using $3a_I + 1 = c_j(3a_K+1) + (3d_j - c_j + 1)$, yields the update rule

$$c_{j+1} = c_j \cdot 2^{v_K^{(j)} - v_I^{(j)}}, \qquad d_{j+1} = \frac{3d_j - c_j + 1}{2^{v_I^{(j)}}}.$$

(The numerator changes sign relative to the T_- rule.) We define the T_+-analogue of the X-invariant by

$$X_{\mathrm{end}}^{+}(r, L) := c_J - 3d_J,$$

where J is the termination index. An odd r with $v_2(r+1) \geq 1$ and $L > v_2(r+1)$ is a *T_+-obstruction residue at level L* if $X_{\mathrm{end}}^{+}(r, L) = 1$; we write $\mathcal{O}_L^{+} \subseteq \{1, \ldots, 2^L\}$ for the set of such residues.

Remark 7.1. The sign convention $X_{\mathrm{end}}^{+} := c - 3d$ (rather than $3d + c$) is the one forced by the affine algebra: with this combination the analogue of the parity lemma (Lemma 2.5) yields the shift-s normal form $(c_J, d_J) = (4^s, (4^s-1)/3)$, and the residue-level bijection of Theorem 7.3 below is exact.

7.2. **The residue involution.**

Lemma 7.2 (Residue-level involution). *Let r be odd with $v := v_2(r-1) \geq 1$ and $L > v$, and set $\overline{r} := 2^L - r$. In the conclusions below, write $V_K^{(j)} := \sum_{i<j} v_K^{(i)}$ and $V_I^{(j)} := \sum_{i<j} v_I^{(i)}$ for the cumulative valuations along the two tracks; the proof shows that they coincide across the T_+ and T_- reductions, so a single notation suffices. Conclusions:*

(i) $\overline{r}$ is odd with $v_2(\overline{r}+1) = v < L$ (so the T_+ reduction at $\overline{r}$ in the sense of Section 7 uses the same parameter v);

(ii) the T_- parallel reduction at (r, L) and the T_+ parallel reduction at $(\overline{r}, L)$, both with parameter v, terminate at the same index J;

(iii) for every $0 \leq j \leq J$,

$$a_K^{+,(j)} \equiv -\,a_K^{-,(j)} \pmod{2^{L-V_K^{(j)}}}, \quad a_I^{+,(j)} \equiv -\,a_I^{-,(j)} \pmod{2^{L-v-V_I^{(j)}}}, \tag{4}$$

and $v_K^{+,(j)} = v_K^{-,(j)}$, $v_I^{+,(j)} = v_I^{-,(j)}$ whenever $j < J$;

(iv) the affine data satisfy

$$(c_j^+, d_j^+) = (c_j^-,\, -d_j^-) \qquad (0 \leq j \leq J). \tag{5}$$

Proof. (i): $\overline{r} + 1 = 2^L - (r-1)$ has, by the ultrametric identity with $v_2(r-1) = v < L$, 2-adic valuation exactly v. Oddness of $\overline{r}$ is immediate.

(iii) and (ii) jointly: we prove the residue congruences in (4) for $j = 0, 1, \ldots$ by induction, and deduce simultaneous termination and the valuation equalities along the way.

Base case ($j = 0$). K-track: $a_K^{+,(0)} = \overline{r} = 2^L - r = -a_K^{-,(0)} + 2^L$. I-track: $a_I^{-,(0)} = (r-1)/2^v$ and $a_I^{+,(0)} = (\overline{r}+1)/2^v = (2^L-(r-1))/2^v = 2^{L-v} - a_I^{-,(0)}$. Both congruences in (4) hold at $j = 0$. For the valuation equality at $j = 0$, suppose the T_- reduction is still running, i.e. $v_K^{-,(0)} < L$. Then $3\overline{r} + 1 = 3 \cdot 2^L - (3r-1)$ has summands of valuations L and $v_K^{-,(0)}$ respectively, with $v_K^{-,(0)} < L$; the ultrametric identity gives $v_K^{+,(0)} = v_K^{-,(0)}$. The analogous calculation on the I-track gives $v_I^{+,(0)} = v_I^{-,(0)}$.

Inductive step. Assume (4) at index j and that both reductions have not yet terminated at index j (so $v_K^{(j)} < L - V_K^{(j)}$ and $v_I^{(j)} < L - v - V_I^{(j)}$, with valuations equal across $+/-$ by the inductive valuation equalities). For the K-track,

$$3a_K^{+,(j)} + 1 \equiv 3(-a_K^{-,(j)}) + 1 = -(3a_K^{-,(j)} - 1) \pmod{2^{L-V_K^{(j)}}}.$$

The right-hand side has valuation $v_K^{(j)} < L - V_K^{(j)}$, so the left-hand side has the same valuation $v_K^{+,(j)} = v_K^{(j)}$, and dividing through by $2^{v_K^{(j)}}$ gives

$$a_K^{+,(j+1)} = \frac{3a_K^{+,(j)} + 1}{2^{v_K^{(j)}}} \equiv -\,\frac{3a_K^{-,(j)} - 1}{2^{v_K^{(j)}}} = -\,a_K^{-,(j+1)} \pmod{2^{L-V_K^{(j+1)}}}.$$

The same calculation on the I-track gives the analogous congruence and $v_I^{+,(j)} = v_I^{(j)}$.

Simultaneous termination. The termination condition of the $(\overline{r}, L)$ reduction at index j is $v_K^{+,(j)} \geq L - V_K^{(j)}$ or $v_I^{+,(j)} \geq L - v - V_I^{(j)}$. We show termination at exactly J in two directions. *No earlier stop:* if the T_+ reduction stopped at some $j' < J$, then by the inductive step at j' (valid because both reductions still run at j') the valuation equalities $v_K^{+,(j')} = v_K^{-,(j')}$ and $v_I^{+,(j')} = v_I^{-,(j')}$ hold, so the T_- reduction would also satisfy its stop condition at j', contradicting the definition of J. *Stop at J:* the residue congruence (4) at $j = J$ follows by applying the inductive step one final time (which is valid because the inductive hypothesis at $j = J-1$ only requires the reductions to be running there, not at J). Combining this congruence with the T_- K-track stop $3a_K^{-,(J)} - 1 \equiv 0 \pmod{2^{L-V_K^{(J)}}}$ yields $3a_K^{+,(J)} + 1 \equiv -(3a_K^{-,(J)} - 1) \equiv 0 \pmod{2^{L-V_K^{(J)}}}$, so $v_K^{+,(J)} \geq L - V_K^{(J)}$ and the T_+ K-track stops at J; the analogous calculation on the I-track gives $v_I^{+,(J)} \geq L - v - V_I^{(J)}$.

(iv): At $j = 0$, $c_0^+ = 2^{-v} = c_0^-$ and $d_0^+ = +2^{-v} = -\,d_0^-$. Inductively, given $(c_j^+, d_j^+) = (c_j^-, -d_j^-)$ and the valuation equalities $v_K^{+,(j)} = v_K^{-,(j)}$, $v_I^{+,(j)} = v_I^{-,(j)}$,

$$\begin{aligned} c_{j+1}^+ &= c_j^+ \cdot 2^{v_K^{(j)} - v_I^{(j)}} = c_{j+1}^-, \\ d_{j+1}^+ &= \frac{3d_j^+ - c_j^+ + 1}{2^{v_I^{(j)}}} = -\,\frac{3d_j^- + c_j^- - 1}{2^{v_I^{(j)}}} = -\,d_{j+1}^-. \end{aligned}$$ □

7.3. The bijection theorem.

Theorem 7.3. *For every level L, the involution $r \mapsto \overline{r} := 2^L - r$ restricts to a bijection $\mathcal{O}_L \leftrightarrow \mathcal{O}_L^+$. Consequently $|\mathcal{O}_L^+| = |\mathcal{O}_L|$, and Theorems 1.1 to 1.3 hold verbatim for $\mathcal{O}^+$, with the same density constant c_W and factor complexity function $p_W^+(\ell) = 2^\ell$.*

Proof. For r odd with $v_2(r-1) \geq 1$ and $L > v_2(r-1)$, equation (5) of Lemma 7.2(iv) at the terminal index J together with the sign convention $X_{\text{end}}^+ := c - 3d$ of Remark 7.1 gives

$$X_{\text{end}}^+(\overline{r}, L) = c_J^+ - 3d_J^+ = c_J^- - 3 \cdot (-d_J^-) = c_J^- + 3d_J^- = X_{\text{end}}(r, L).$$

Hence $r \in \mathcal{O}_L$ iff $\overline{r} \in \mathcal{O}_L^+$. Since $r \mapsto \overline{r}$ is an involution on the odd residues in $\{1, \ldots, 2^L - 1\}$, it restricts to a bijection $\mathcal{O}_L \leftrightarrow \mathcal{O}_L^+$. Concretely, at $L = 5$ this gives $\mathcal{O}_5^+ = \{5\}$, and at $L = 6$ it gives $\mathcal{O}_6^+ = \{2^6 - r : r \in \mathcal{O}_6\} = \{5, 37, 45\}$, the anchor set parallel to $\mathcal{O}_6$ in the T_+ analogue of the factor-complexity construction below.

For the remaining claims, observe that Lemma 7.2 conjugates each T_- basic step into the corresponding T_+ basic step. The T_+ lift theorem (the analogue of Theorem 1.1) follows by composing the involution with the T_- lift: given $\overline{r_0} \in \mathcal{O}_{L_0}^+$, set $r_0 := 2^{L_0} - \overline{r_0} \in \mathcal{O}_{L_0}$; by Theorem 1.1, both

r_0 and $r_0 + 2^{L_0}$ lie in $\mathcal{O}_{L_0+1}$, and their level-(L_0+1) involution images $2^{L_0+1} - r_0 = \overline{r_0} + 2^{L_0}$ and $2^{L_0+1} - (r_0 + 2^{L_0}) = \overline{r_0}$ lie in $\mathcal{O}^+_{L_0+1}$ by the level-(L_0+1) version of the bijection. Hence both T_+-lifts of $\overline{r_0}$ are again T_+-obstructions. Positive density and full factor complexity for $\mathcal{O}^+$ then propagate exactly as in Sections 4 and 5, with $\mathcal{O}^+$ in place of $\mathcal{O}$ throughout the argument. □

8. Open questions

8.1. **The exact value of c_W.** The principal open question is to determine c_W exactly, or equivalently to give a quantitative bound on the growth of $|\mathcal{A}_L^{G\neq 0}|$. Three approaches present themselves, each requiring substantial additional work.

(i) Substitution-theoretic. Stérin [Sté20] realizes T_+ as a four-state finite-state transducer between bases 2 and 3. An analogous construction for the parallel reduction, combined with Pansiot's classification [Pan84] (refined by Devyatov [Dev15]), would determine the asymptotic growth class of $\mathcal{A}_L^{G\neq 0}$. The relevant question concerns *non-primitive* substitution shifts: the primitive case is ruled out by Corollary 5.2, so any realisation must lie outside that classification (Remark 5.3). The natural setting for such a realisation is the S-adic framework [BD14].

(ii) Markov-operator-spectral. Wirsching [Wir98] constructs a Markov operator W_3 on residue-density functions for the T_+ predecessor problem. An analogous operator for the parallel reduction would have an action on (c,d)-space; its spectrum would determine the asymptotic growth of the shift-class populations $|\mathcal{O}_L^{G_s}|$. A unified treatment — for instance, expressing c_W in terms of Wirsching's invariant density φ — appears not to be in the literature.

(iii) Generating-function-combinatorial. A direct enumeration via generating functions in L is in principle accessible, but the absence of a closed-form recursion — the count depends on the joint distribution of stopping times and shift indices — makes the analysis delicate.

8.2. **Nontrivial cycles.** The shift-zero class $\mathcal{O}_L^{G_0}$ encodes residues for which the two parallel stopping times remain rigidly tied. Whether the existence (or absence) of additional nontrivial T_--cycles is reflected in the structure of $\mathcal{O}_L^{G_0}$ for large L — in particular, whether one can extract a Diophantine obstruction from the strict lift balance of Corollary 6.4 — is a natural and apparently open question. Classical lower bounds on the length of nontrivial T_+-cycles, due to Eliahou [Eli93] and extended by Belaga and Mignotte [BM06], indicate how rigidly the Diophantine geometry constrains cycle structure; an analogous bound in terms of $\mathcal{O}_L^{G_0}$ would be of independent interest.

Extension to the $(3n + b)$ family

The constructive lift, the positive-density bound, and the full-factor-complexity result extend to the family $T_b(n) := (3n + b)/2^{v_2(3n+b)}$ for every odd b. The bridge is a multiplicative residue-bijection on $\{1, \ldots, 2^L\}$ that conjugates the obstruction structure across the family and preserves the shift index; the involution $r \mapsto 2^L - r$ used in Section 7 is a special case. A detailed treatment is in preparation as a separate paper.

Further extensions

We have also established significant partial results for general multiplier a — in particular structure theorems for $a = -1$ and for the negative-Mersenne family $a = -(2^v - 1)$ with explicit lower bounds on c_W. A complete treatment of the $(an+b)$ family raises new phenomena (Mersenne asymmetry, a sub-class hierarchy, a signed sync index) and is the subject of ongoing work.

Funding and competing interests

The author received no funding for this research and declares no competing interests.

Appendix A. Computational verification

The algorithmic X-invariant criterion of Section 2 is implemented in Python in the supporting code that accompanies this paper, collected in the directory `code/python/` alongside the manuscript source and mirrored at https://github.com/yaccob/collatz/tree/main/manuscripts/obstruction_residues/code. A permanent archival snapshot is deposited on Zenodo under the concept DOI 10.5281/zenodo.20356496, which resolves to the most recent archived version. The directory contains the verification scripts for each rigorous result (lift theorem, atom decomposition, the no-shift-zero-atom theorem (Theorem 6.2), full factor complexity, simultaneous stop) together with a `README.md` mapping each script to the result it certifies. For each level $L \le L_{\max}$, the relevant script enumerates all odd residues r with $v_2(r - 1) \ge 1$, runs the parallel reduction, and records the terminal data. The set $\mathcal{O}_L$ is read off as the preimage of $\{(c, d) : 3d + c = 1\}$.

We have run this enumeration through $L = 16$ in Python, producing the counts of Table 3. A second, independent Rust implementation (`code/rust/`) reproduces these counts for $L = 5, \ldots, 16$ and extends the rigorous density bounds of Table 1 to $L = 32$.

The data satisfies the strict lift balance $|\mathcal{O}_L| = 2|\mathcal{O}_{L-1}| + |\mathcal{A}_L^{G\neq 0}|$ of Corollary 6.4 at every level, and is consistent with Theorems 1.1, 6.2 and 6.5 (series convergence).

The rigorous bound $c_W \ge 7556/65536 \approx 0.115$ of Theorem 1.2 is conditional on the correctness of this enumeration. The check is self-validating in a useful sense: the strict lift balance of Corollary 6.4 is asserted row-by-row by the script `count_obstructions.py` against the previous-level count, so

L	$\lvert\mathcal{O}_L\rvert$	$\lvert\mathcal{O}_L^{G_0}\rvert$	$\lvert\mathcal{O}_L^{G\neq 0}\rvert$	$\lvert\mathcal{A}_L^{G\neq 0}\rvert$
5	1	0	1	1
6	3	1	2	1
7	8	4	4	2
8	19	12	7	3
9	42	31	11	4
10	91	73	18	7
11	194	164	30	12
12	409	358	51	21
13	855	767	88	37
14	1776	1622	154	66
15	3671	3398	273	119
16	7556	7069	487	214

TABLE 3. Direct enumeration counts at levels $L = 5, \ldots, 16$: total obstructions, shift-zero and shift-nonzero subclasses, and atomic shift-nonzero obstructions. The decomposition $|\mathcal{O}_L| = |\mathcal{O}_L^{G_0}| + |\mathcal{O}_L^{G\neq 0}|$ is read off each row, and the strict lift balance $|\mathcal{O}_L| = 2|\mathcal{O}_{L-1}| + |\mathcal{A}_L^{G\neq 0}|$ (Corollary 6.4) holds row-by-row against the previous one for $L \geq 6$.

any drift in the enumeration would manifest as an immediate hard failure rather than silently shifting the bound. Each rigorous-verification script exits non-zero on any violation of the asserted theorem (a convention checked statically by `meta_test_exit_convention.py`), so the bound is reproducible end-to-end by running the script.

References

[AS03] Jean-Paul Allouche and Jeffrey Shallit, *Automatic sequences: Theory, applications, generalizations*, Cambridge University Press, 2003.

[BD14] Valérie Berthé and Vincent Delecroix, *Beyond substitutive dynamical systems: S-adic expansions*, RIMS Kôkyûroku Bessatsu **B46** (2014), 81–123.

[BM06] Edward G. Belaga and Maurice Mignotte, *Walking cautiously into the Collatz wilderness: Algorithmically, number-theoretically, randomly*, Fourth Colloquium on Mathematics and Computer Science: Algorithms, Trees, Combinatorics and Probabilities, Discrete Mathematics and Theoretical Computer Science Proceedings AG, 2006, pp. 249–260.

[BR10] Valérie Berthé and Michel Rigo (eds.), *Combinatorics, automata and number theory*, Encyclopedia of Mathematics and its Applications, vol. 135, Cambridge University Press, 2010.

[Cas97] Julien Cassaigne, *Complexity and special factors*, Bulletin of the Belgian Mathematical Society — Simon Stevin **4** (1997), no. 1, 67–88.

[Cra78] R. E. Crandall, *On the "$3x+1''$ problem*, Mathematics of Computation **32** (1978), no. 144, 1281–1292.

[Dev15] Rostislav Devyatov, *On the subword complexity of morphic words*, Acta Informatica **52** (2015), no. 4, 379–425.

[DHS99] Fabien Durand, Bernard Host, and Christian Skau, *Substitutional dynamical systems, Bratteli diagrams and dimension groups*, Ergodic Theory and Dynamical Systems **19** (1999), no. 4, 953–993.

[Eli93] Shalom Eliahou, *The* $3x+1$ *problem: new lower bounds on nontrivial cycle lengths*, Discrete Mathematics **118** (1993), 45–56.

[Eve77] C. J. Everett, *Iteration of the number-theoretic function* $f(2n) = n$, $f(2n+1) = 3n+2$, Advances in Mathematics **25** (1977), no. 1, 42–45.

[Fog02] N. Pytheas Fogg, *Substitutions in dynamics, arithmetics and combinatorics*, Lecture Notes in Mathematics, vol. 1794, Springer, 2002.

[Kor94] Ivan Korec, *A density estimate for the* $3x+1$ *problem*, Mathematica Slovaca **44** (1994), no. 1, 85–89. MR 1290275

[Lag85] Jeffrey C. Lagarias, *The* $3x+1$ *problem and its generalizations*, American Mathematical Monthly **92** (1985), 3–23.

[Lag10a] ———, *The* $3x+1$ *problem: an annotated bibliography (1963–1999) and (2000–2009)*, 2010, arXiv:math/0309224 and arXiv:math/0608208.

[Lag10b] Jeffrey C. Lagarias (ed.), *The Ultimate Challenge: The* $3x+1$ *Problem*, American Mathematical Society, 2010.

[LM95] Douglas Lind and Brian Marcus, *An introduction to symbolic dynamics and coding*, Cambridge University Press, 1995.

[Lot02] M. Lothaire, *Algebraic combinatorics on words*, Encyclopedia of Mathematics and its Applications, vol. 90, Cambridge University Press, 2002.

[MH38] Marston Morse and Gustav A. Hedlund, *Symbolic dynamics*, American Journal of Mathematics **60** (1938), no. 4, 815–866.

[MH40] ———, *Symbolic dynamics II. Sturmian trajectories*, American Journal of Mathematics **62** (1940), no. 1, 1–42.

[Pan84] Jean-Jacques Pansiot, *Complexité des facteurs des mots infinis engendrés par morphismes itérés*, Automata, Languages and Programming (ICALP 1984), Lecture Notes in Computer Science, vol. 172, Springer, 1984, pp. 380–389.

[Sté20] Tristan Stérin, *The Collatz process embeds a base conversion algorithm*, 2020.

[Tao22] Terence Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum of Mathematics, Pi **10** (2022), Paper No. e12.

[Ter76] Riho Terras, *A stopping time problem on the positive integers*, Acta Arithmetica **30** (1976), no. 3, 241–252.

[Wir98] Günther J. Wirsching, *The dynamical system generated by the* $3n+1$ *function*, Lecture Notes in Mathematics, vol. 1681, Springer, 1998.

Independent researcher, Vienna, Austria
Email address: `jakob.stemberger@yaccob.net`
URL: `https://orcid.org/0009-0001-2746-6877`